

An Adaptive Low-Light Image Enhancement Using Canonical Correlation Analysis

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Abstract—Images captured under low-illumination conditions exhibit numerous unwanted appearances such as color distortion, low brightness, a narrow gray range, low contrast, and considerable noise. An excellent low-light image enhancement framework overcomes all undesirable attributes of low-light images. Here, a new fusion technique is introduced based on the canonical correlation analysis of two proposed adaptive Retinex and beta-hyperbolic secant distribution (BHSD) models. First, an adaptive enhanced reflectance image is obtained by a new weighted multiscale Gaussian function and proposed adaptive reflectance map. For the second enhanced image, logarithmic image processing is modified according to the illuminance's factor, and the exponential function is incorporated for contrast boosting. The error function is used for the modified cumulative distribution function of BHSD to obtain the second enhanced image. The proposed framework is compared with state-of-the-art low-light methods for visual and parametric comparison. The results show that the designed framework produces the best results in terms of high contrast, natural brightness, vivid color combination, perfect structure, and texture information with preserving naturalness. Among all comparable methods, the designed technique produces a good result in terms of rich scene details and color reproduction in dark regions.

Index Terms—Adaptive reflection enhanced image, beta-hyperbolic secant distribution, canonical correlation analysis, low-light image enhancement, multiscale Gaussian function.

I. INTRODUCTION

TODAY photo plays a crucial role in daily life activities because people widely use photography equipment to record their daily activities. They share beautiful moments on several social media platforms by taking photos. Moreover, the images are being used in several computer vision problems, including image recognition [1], image understanding, and object detection [2]. Image processing systems are also doing great jobs in industrial material recognition by a robot [3], automated driving [1], and disease detection [2]. But these

systems also fail to produce desired or exact results due to various unavoidable factors during the image acquisition process, especially under unfavorable conditions. Low-light conditions such as cloudy weather, indoor lighting, and night-time lighting degrade the quality of the image. In low-light environments, the colors of images are discriminatory, with considerable noise in captured images as well. The reflection of the target surface is also feeble, making a dark image. These severe degradations in image quality prevent visual systems from achieving their desired targets. Low-light Image enhancement approaches can also solve problems and reduce the reliability of photography equipment. Low-light image enhancement approach [1] comes under the preprocessing step of industrial applications for intelligent driving, object pose measurement, detect vehicles and pedestrians, and image synthesis. It has very significant roles and applications in the domains of robots, industrial production, aerospace, vehicles, rotorcraft, and ocean [4].

Several low-light image enhancement approaches [5], [6], [7] have been introduced in the literature, such as histogram equalization (HE) based approaches, methods based on the Retinex model, wavelet domain-based techniques, and fusion-based algorithms. Among all methods, the Retinex concept [8], [9] works better for low illuminance images because it is based on the concept of human visual perception, especially for scenes that contain both dark and bright areas. It decomposes the image into illuminance and reflectance components in which a bright channel is mainly preferred to obtain the illuminance map. Guo et al. [10] proposed a method that optimized the initial illuminated image's refinement, and the gamma correction approach was explored to achieve the enhancement. Still, this technique generates over-enhanced images along with noises. Primarily, in Retinex theory-based frameworks, the final enhanced image is obtained by the fusion operation where more than one enhanced image is combined by several fusion approaches [11], [12], [13], [14] like principal component analysis (PCA) and Gaussian and Laplacian pyramids technique. In [12], the PCA approach combines the two enhanced images to obtain final low-light enhanced results. Singh and Bhandari [13] presented an algorithm that is an extended version of the PCA-based framework [12]. But enhanced images contain hole artefacts and noises that make unnatural brightness and displeasing visuals.

The abovementioned approaches obtain good low-quality images compared to the traditional Retinex techniques [9]. However, there is still room for improvement in edge details, color information, hole artefact, and natural brightness. The overall tonality of low-light enhanced images has been flatter, while the

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improved images generate a stronger contrast with artefacts. The overenhancement problems also occur in these enhanced images with unnatural color boosting. Generally, in the low light image enhancement methods, authors have mainly concentrated on the lower range pixel values of the images because of the lack of illumination and the intensity difference of the corresponding pixels among all three red, green, and blue channels of the color image are limited.

In this article, the proposed adaptive image enhancement using the canonical correlation analysis (AECCA) framework focuses on enhancing color, boosting the edge details, contrast diversity, and brightness vibrancy. The proposed AECCA produces more color deviations with more visual details for darker and brighter image regions because it proposed two different adaptive frameworks based on the Retinex model and beta-hyperbolic secant distribution (BHSD). Consequently, it is easy to distinguish color and edge information with the natural brightness of the image. It explores a new fusion technique combining enhanced images based on canonical correlation analysis (CCA). The proposed approach does not produce such artefacts in terms of unnatural enhancements like comparable techniques generated; artefacts seriously degrade the visual quality, which restricts the functionality of various visual systems. The high-lighted contributions of this article are mentioned as follows.

- 1) The proposed weighted multiscale Gaussian function adequately meets the requirements of illumination component extraction in terms of structure information.
- 2) A new adaptive reflectance-enhanced framework is proposed that is very simple and effective for weakly illuminated images.
- 3) Afterwards, a modified S-curve function based on the BHSD is introduced to boost the overall brightness and contrast of the low-light image.
- 4) A new CCA-based fusion technique is presented that fuses two enhanced images using the concept of the projection technique.

The rest of this article is organized as follows. Section II contains the proposed methodology, the AECCA framework for the weakly illuminated image enhancement problem. Section III discusses the performance of methods in quantitative and qualitative assessments. Finally, Section IV concludes this article.

II. PROPOSED METHOD

In this article, low-light color images I_{IN} are enhanced using the Retinex model and hyperbolic function. The Retinex model proposes a unique multiscale Gaussian function to decompose the image into reflectance and illuminance elements. This illumination map is utilized to generate the first enhanced image by a novel adaptive reflectance approach. Before implementing Retinex theory, the red, green, and blue (RGB) color image is changed into the hue, saturation, and value (HSV) space, and all processes are performed in only the V-layer.

The initial step is to find the illuminance and adaptive enhanced reflectance components of the V-layer, where the considered range of intensity values is 0 to 255 for the $I_V(m, n)$ image. A unique algorithm obtains the second enhanced image

based on the cumulative distribution function of BHSD (CDF-BHSD) [15], where illuminance's factor is explored for adaptive image enhancement. The final enhanced V-layer is achieved by a novel fusion algorithm where the canonical correlation technique is the first time explored to combine two images using their histograms. After obtaining the enhanced V, HSV space is converted back to RGB space, and enhanced color output is obtained I_{OUT} , as shown in Fig. 1.

A. Estimation of the Proposed Illumination Map

Here, the Retinex model [11], [16] separates an image $I(m, n)$ into reflectance $R(m, n)$ and illuminance $L(m, n)$ maps, as mentioned in (1). In (1), R deals with the detailed reflecting information about the object itself. L signifies the reflecting intensity of ambient light. The texture details of the image are captured by the reflectance, whereas structure information is illustrated in the illuminance component. The Retinex model decomposes the image into structural and texture layers [17]

$$I(m, n) = L(m, n) \bullet R(m, n) \quad (1)$$

where $\bullet$ denotes the element multiplication operator, this article introduces a unique framework to estimate a low-light scene's exact and accurate illuminance component $L_V(m, n)$. It consists of a weighted multiscale Gaussian function $G(m, n)$ in which more weight is assigned to the smaller scale parameter (σ) compared to the higher scale factor. A low-light image as shown in Fig. 2(a), and its illuminance maps are represented in Fig. 2, where the proposed illuminance in Fig. 2(c) is better than the conventional averaging approach, as shown in Fig. 2(b). The proposed weighted multiscale Gaussian function extracts the structural edges in the illuminance image better than the traditional method-based illuminance map, as shown in Fig. 2. It concludes that the presented approach can accurately extract the illuminance map of the image to structure details. So, the texture details of the image can be well estimated in the designed adaptive reflectance function.

In Gaussian function methodology, the high compression of the dynamic range and apparent visual characteristic of the image for the smaller scale parameter value [8], [9]. More weight is allotted to the illuminance image for the perfect estimation of illuminance values, which has a smaller scale factor than the larger parameter-based illuminance images. The proposed illuminance component is extracted by the weighted multiscale Gaussian functions as follows:

$$L_V(m, n) = \sum_{i=1}^N W_i(I_V(m, n) * G_i(m, n)) \quad (2)$$

where $*$ is the convolution operator and W_i denotes the weight coefficient of 0.5, 0.3, and 0.2 for illuminance images at different scale parameters ($\sigma = 15, 80$, and 250) [9], respectively. The N represents the number of scales used which is 3. The Gaussian function is implemented to obtain the illuminance map of the low-light image, which is represented in

$$G(m, n) = F_{norm} \exp \left(-(m^2 + n^2) / \sigma \right). \quad (3)$$

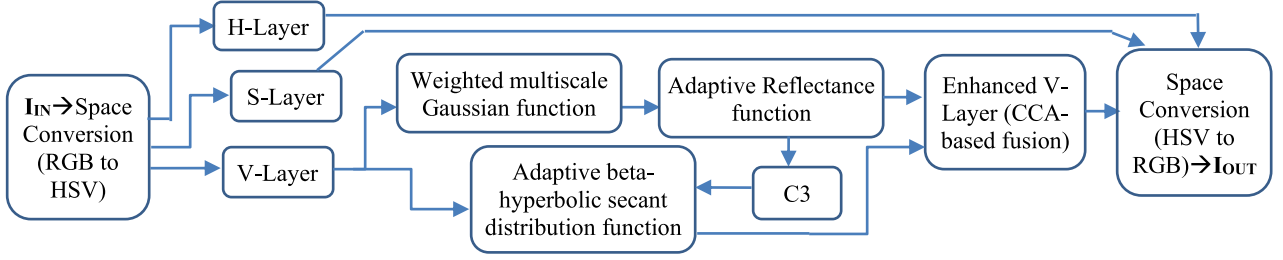


Fig. 1. Flowchart of the proposed adaptive low-light image enhancement using canonical correlation analysis (AECCA) framework.

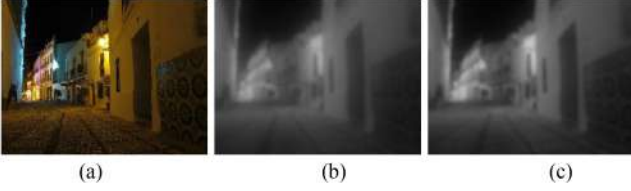


Fig. 2. (a) Input image and its illuminance maps. (b) Traditional multiscale Gaussian approach. (c) Proposed weighted multiscale Gaussian approach.

The value of normalization factor F_{norm} is to be set to satisfy the Gaussian function condition, i.e., $\iint G(m, n) = 1$. To estimate the illumination image, the convolution operation is performed between image $I_V(m, n)$ and Gaussian function $G(m, n)$.

B. Adaptive Reflectance-Based Enhancement

After extracting the illuminance element of the low-light image, an adaptive reflectance function is proposed to obtain the enhanced image. According to the traditional Retinex model, the reflectance image is estimated by (1), but in this article, a new algorithm is proposed to achieve a better-enhanced image, as shown in Fig. 2. The enhanced image I_1 is obtained using the designed normalized reflectance function R_{adapt} as follows:

$$I_1(m, n) = R_{\text{adapt}}(m, n) / \max(R_{\text{adapt}}(m, n)). \quad (4)$$

First, compute the I_{max} image that contains the maximum intensity value of two images at each coordinate (m, n) where illuminance L_V and V-layer I_V images are employed for this purpose as represented in

$$I_{\text{max}}(m, n) = \max(L_V, I_V). \quad (5)$$

The next step is to find the mean values of I_{max} and I_V images that are m_1 , and m_2 , respectively; these mean intensities are used to set the constant $C1$, $C2$, and $C3$ (illuminance's factor) values for the proposed adaptive reflectance function R_{adapt}

$$\begin{aligned} R_{\text{adapt}}(m, n) &= [C1 \times I_V(m, n)] / [I_{\text{max}}(m, n) + C2]. \\ \text{if } m_1 &\geq m_2 \\ C1 &= m_1, C2 = m_2, \text{ and } C3 = \sqrt{m_1/m_2} \\ \text{else} \\ C1 &= m_2, C2 = m_1, \text{ and } C3 = \sqrt{m_2/m_1} \\ \text{end.} \end{aligned} \quad (6)$$

Here, all constant values are adaptive in nature because they depend on the image's feature, and the modified reflectance component is based on the $C1$ and $C2$. The $C3$ represents the illuminance factor of a low-light image.

C. Enhancement is Based on Adaptive Beta-Hyperbolic Secant Distribution

This article proposes an algorithm based on a newly designed CDF-BHSD [15] that is I_{erf} . For the high contrast enhancement and to control the enhancement rate, a normalization operation is performed on the I_{erf} image to linearly scale the intensity values to be suitable for the complete dynamic range of grayscale values. The normalization function is calculated as follows:

$$I_2(m, n) = (I_{\text{erf}} - \min(I_{\text{erf}})) / (\max(I_{\text{erf}}) - \min(I_{\text{erf}})). \quad (7)$$

In the fields of probability and statistics, the standard BHSD is a well-known algorithm, and the CDF of this function follows the S-curve properties that improve the contrast and brightness of low-light images. The primary purpose of this designed framework is to sustain the natural illumination in the bright regions by controlling being extremely enhanced while improving the brightness and contrast in the dark areas. Hence, the enhanced image consists of proper image details and natural brightness. The proposed algorithm explores the properties of both logarithmic and exponential functions in a unique way to improve the contrast of the areas which contain low- and mid-intensities and properly enhance high-intensities regions. The designed logarithmic scaling function is defined in

$$I_{\log}(m, n) = \frac{\max(I_V)}{\log(\max(I_V + C3))} \times \log(I_V + C3). \quad (8)$$

Next, the V-layer image $I_V(m, n)$ is used again by a noncomplex exponential function to improve the contrast of an image and this exponential operation is performed using [16]

$$I_{\exp}(m, n) = 1 - \exp(-I_V). \quad (9)$$

The logarithmic image processing technique [18] is explored to combine both $I_{\log}$ and $I_{\exp}$ images to reach the best visual enhancement of the low-light image. This model generates I_{LE} image as expressed in (10). After that, an adaptive CDF-BHSD is contrasted with boosting the overall contrast and brightness of the low-light image. In general, the CDF of this algorithm is one kind of S-curve function that adjusts the contrast and brightness. The proposed adaptive CDF-BHSD B_{Hyper} is mentioned in (11)

as follows:

$$I_{LE}(m, n) = (I_{\log} + I_{\exp}) / (1 + I_{\log} \times I_{\exp}) \quad (10)$$

$$I_{erf}(m, n) = erf(B_{Hyper}) = erf(C3 \times \arctan(\exp(I_{LE}))). \quad (11)$$

The error function (*erf*) [19] is mainly used to increase the curvy transformation of the CDF-BHSD process because this operator is beneficial in increasing the brightness of the dark areas of low-light images. Usually, the standard CDF-BHSD [15] uses constant value that is $\pi/2$. In this article, the constant value $C3$ is used in the place of $\pi/2$ to adaptively control the amount of enhancement because it directly depends on the V-layer and illuminance images. So, this proposed CDF-BHSD technique helps to preserve the naturalness of the brightness and improve the contrast by regularizing the image tonality. The proposed function makes and appears the enhanced image like the observed natural scene.

D. Image Fusion Using Canonical Correlation Analysis

The final enhanced image is formed by a fusion technique that is based on canonical correlation analysis (CCA) [20], [21]. The CCA is used for the first time to combine two images to produce an enhanced image based on the coefficients of the canonical correlation method. This technique uses the histograms of two images to generate the A_1 and A_2 . The histograms of both I_1 and I_2 enhanced images are $H1$ and $H2$, which are explored to find the weights

$$[A_1, A_2] = \text{canoncorr}(H1, H2). \quad (12)$$

The final enhanced V-layer of the low-light image I_{V_enh} is obtained by (13), where *canoncorr* indicates the canonical correlation operator, and F_1 and F_2 are the fusion weights for the images

$$I_{V_enh}(m, n) = F_1 \times I_1(m, n) + F_2 \times I_2(m, n), \text{ where}$$

$$F_1 = \frac{abs(A_1)}{abs(A_1) + abs(A_2)} \text{ and } F_2 = \frac{abs(A_2)}{abs(A_1) + abs(A_2)}. \quad (13)$$

After obtaining the enhanced V channel image, the color space of the HSV of the enhanced image is converted into the RGB color space, as shown in Fig. 1. The CCA approach measures the correlation between two datasets in which datasets are multidimensional variables. The canonical correlation operator is based on the concept of projections, where one specifies the projected multivariate variable that highly correlates with the other variable for each value separately. In general, the CCA is to maximize the correlation, and the canonical correlation vectors are obtained using a joint covariance of the two variables. The two random variables are $H1 \in \mathbb{R}^{n1}$ and $H2 \in \mathbb{R}^{n2}$, and μ and v are the mean of the $H1$, and $H2$, respectively. The concept is to determine the possible link between two datasets, and the CCA uses the linear combinations based on the linear indices $a^T H1$ and $b^T H2$ of the random vectors. The a and b are the vectors obtained by CCA to maximize the correlation (*corr*) between

two indices $a^T H1$ and $b^T H2$

$$(a^*, b^*) = \arg \max_{a, b} \text{Corr}(a^T H1, b^T H2)$$

$$= \arg \max_{a, b} \frac{a^T \sum_{H1H2} b}{(a^T \sum_{H1H1} a)^{1/2} (b^T \sum_{H2H2} b)^{1/2}} \quad (14)$$

where the covariance (*var*) and cross-covariance (*cor*) are defined for $H1$ and $H2$ random vectors, and they are computed as

$$\text{var}(H1) = \sum_{H1H1} (n1 \times n1) = \sum_{H1H1}$$

$$\text{var}(H2) = \sum_{H2H2} (n2 \times n2) = \sum_{H2H2}$$

$$\text{Cor}(H1, H2) = E(H1 - \mu)(H2 - v)^T$$

$$= \sum_{H1H2}^T (n1 \times n2) = \sum_{H1H2} \quad (15)$$

where E is the expected value, and the objective is invariant to the scaling of a and b ; the projections are restricted to have unit variance

$$(a^*, b^*) = \arg \max_{a^T \sum_{H1H1} a = b^T \sum_{H2H2} b = 1} a^T \sum_{H1H2} b. \quad (16)$$

For this problem, define

$$\lambda = \sum_{H1H1}^{-1/2} \sum_{H1H2}^{-1/2} \sum_{H2H2}. \quad (17)$$

The singular value decomposition technique λ ($n1, n2$) is used to obtain the optimal value, and this matrix is decomposed as

$$\lambda = \Delta \Gamma \Lambda^T, \text{ where } \Delta = (\delta_1, \delta_2, \dots, \delta_k),$$

$$\Gamma = \text{diag}(\gamma_1^{1/2}, \gamma_2^{1/2}, \dots, \gamma_k^{1/2}), \text{ and } \Lambda = (\varepsilon_1, \varepsilon_2, \dots, \varepsilon_k) \quad (18)$$

where $k = \text{rank}(\lambda) = \text{rank}(\sum_{H1H2}) = \text{rank}(\sum_{H2H1})$. δ and ε are the standardized eigenvectors, γ is the nonzero eigenvalues

$$A_1 = \sum_{H1H1}^{-1/2} \delta \text{ and } A_2 = \sum_{H2H2}^{-1/2} \varepsilon \quad (19)$$

where A_1 and A_2 are the canonical correlation vectors and the values of A_1 and A_2 are utilized in (13) to find the weight for fusion technique. The size of both histograms is 256×1 in this article.

III. EXPERIMENTAL RESULTS AND ANALYSIS

The effectiveness of the proposed AECCA framework is evaluated in terms of subjective and objective analysis, where 90 low-light images from four different VV [22], low-light image enhancement (LIME) [10], fusion-based enhancing method (FEM) [11], and DICM [22] datasets are used. All experiments were performed on MATLAB software and the system configurations are Intel (R) Core (TM) i3-5005U CPU @ 2.00 GHz

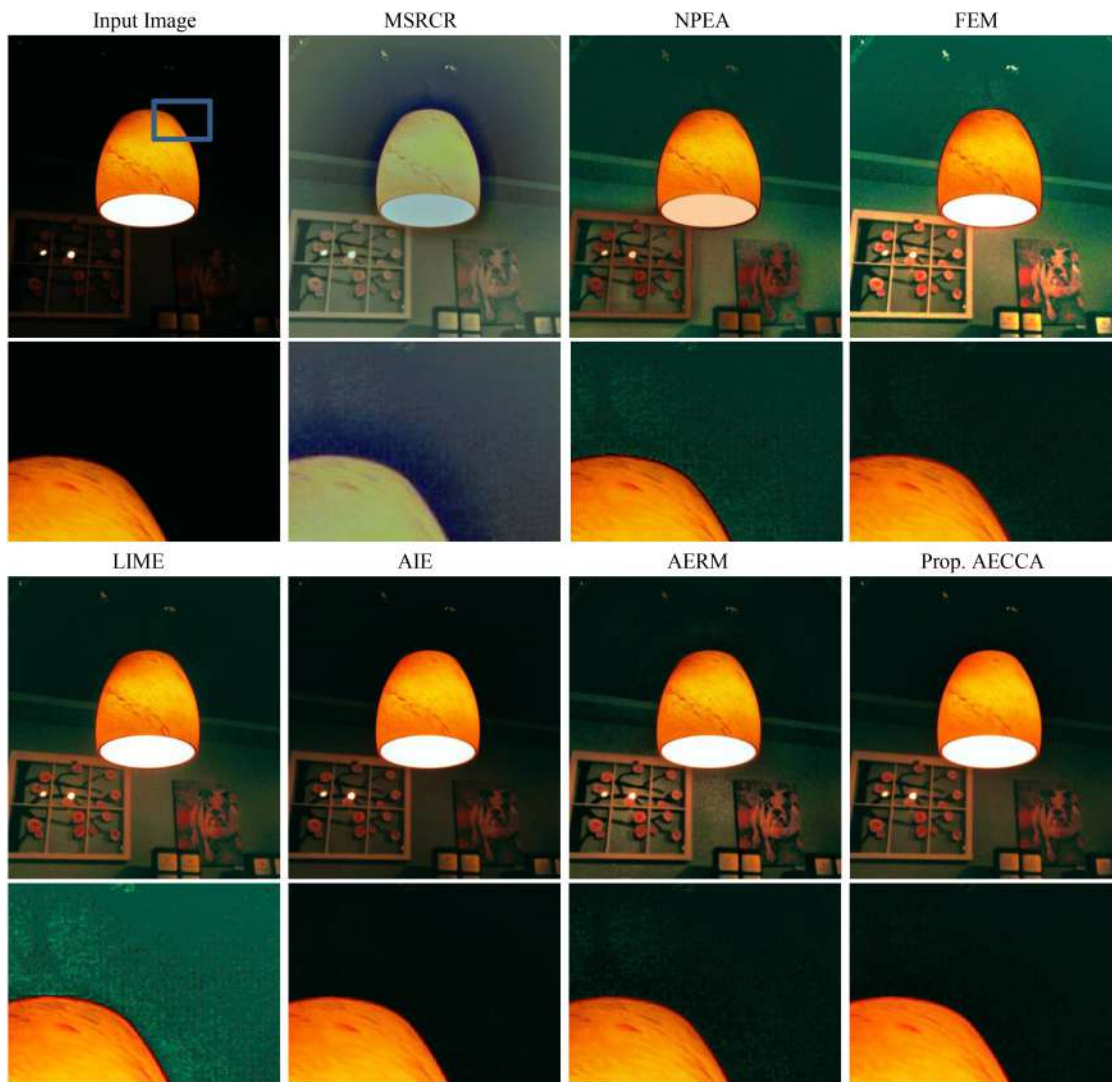


Fig. 3. Input low light image (lamp) and enhanced images after applying low light methods, and the selected zoom portion (indicated in input image) is represented below for each image.

processor and 4 GB RAM. For a fair comparison of the proposed AECCA framework, the well-known low light methods such as multiscale Retinex with color restoration (MSRCR) [9], naturalness preserved enhancement algorithm (NPEA) [23], FEM [11], LIME [10], adaptive image enhancement (AIE) [12], and adaptive enhancement reflection model (AERM) [13] are compared. The average values of the Michelson law measure of enhancement (AME) [24], the second derivative-like measure of enhancement (SDME) [25], the naturalness image quality evaluator (NIQE) [26], and lightness-order-error (LOE) [23] quantitative parameters of all algorithms have been calculated for objective evaluation.

A. Subjective Evaluation

In this section, eight dark images and their enhanced images based on different algorithms are represented in Figs. 3–6. These examples show the visual comparison among all techniques for color images to verify, which method generates the pleasing

visual image for weakly illuminated images. MSRCR and NPEA fail to enhance the right overall images, especially the MSRCR-based images with poor contrast and color distortion. Even it does not preserve the brightness and color of row data. NPEA enhances the dark images but does not change the vivid color and brightness according to image scenes. The main reason for this inadequate improvement is the insufficient enhancement rate that limits the improvement of dark portions of the low-light images, as shown in Figs. 3–6. The overenhancement condition of AERM and LIME makes the appeared color too bright, which causes an unnatural effect. They boosted more color appearance of the images, which created up-and-down effects. The up-and-down impact makes more highlights to dominating pixels and removes the hidden information that makes objects in the image look like statues, as shown in Fig. 5.

At the same time, FEM and AIE produce an inanimate scene effect due to slightly underenhancement conditions and variable color improvement rates. The AIE does not have proper contrast in the weakly illuminated areas of the image, as clearly

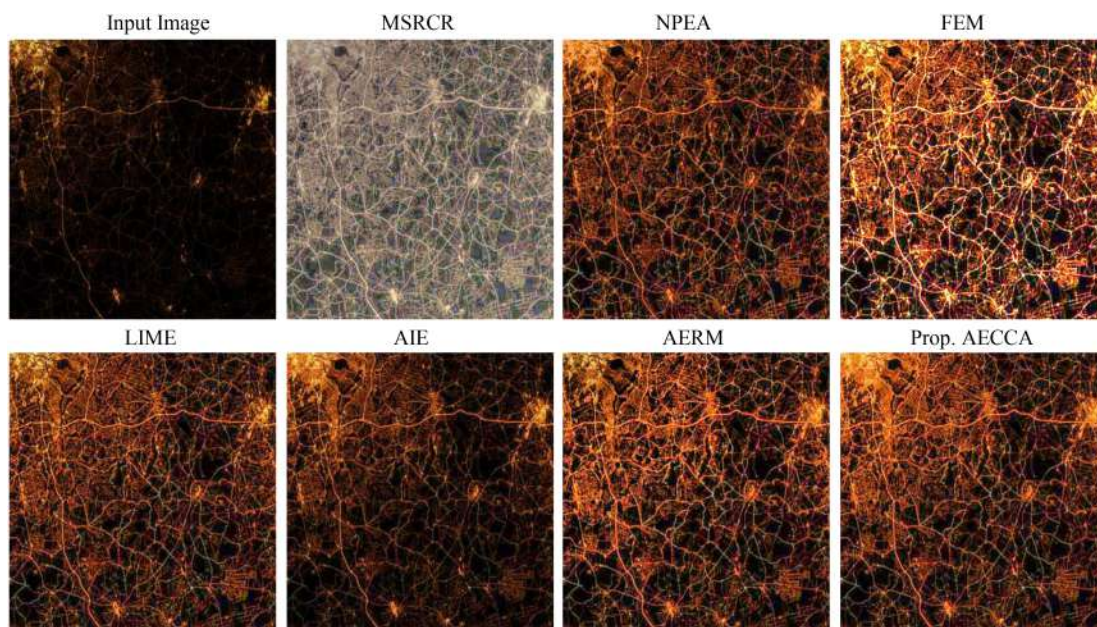


Fig. 4. Input low light image and enhanced images after applying low light image enhancement methods.

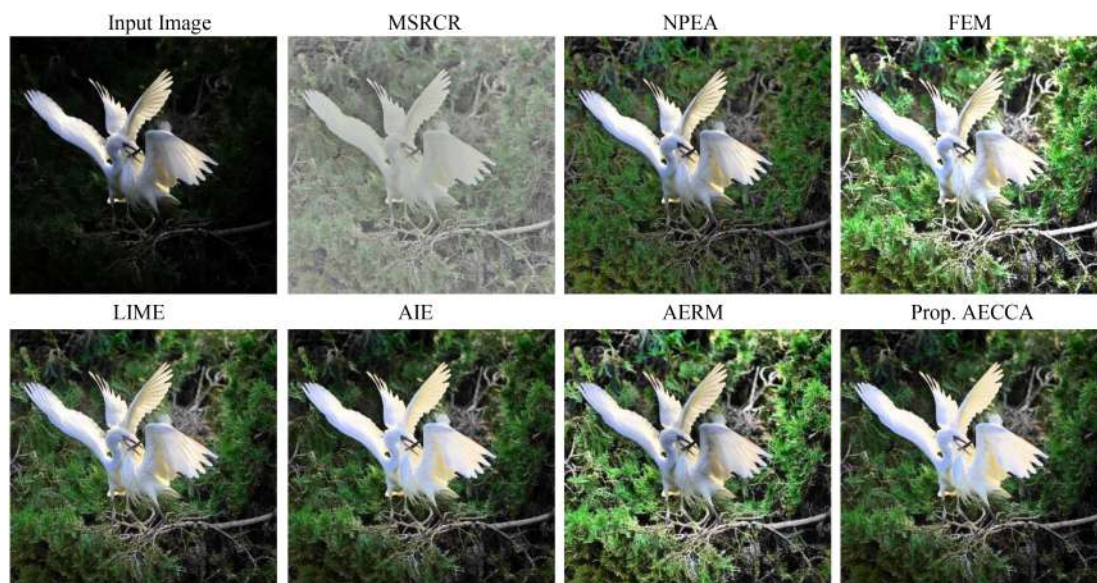


Fig. 5. Input low light image and enhanced images after applying low light image enhancement methods.

demonstrated in Figs. 3–6, where details information is compressed due to low contrast and slightly low brightness. The hidden information is not highlighted accurately, and the appearance of the colors is somewhat dull in the FEM-based images, which can be visualized clearly in Figs. 4–6. The proposed AECCA method generates natural enhancement results without any artefacts, especially on vivid color scenes. High contrast and natural brightness are generated by the presented AECCA algorithm, particularly in all dark regions of images that make the pleasing visual results with more structural details and textual information. In Fig. 3, The zoomed-in local portion in the blue

box (input image) represents that MSRCR, NPEA, MF, LIME, and AERM cause unwanted and halo artefacts around the lamp. Specially MSRCR and LIME even alter the actual scene of the lighting condition due to over enhancement rate.

The AIE generates a decent enhanced image, but it also slightly follows underenhancement condition that makes unnatural color and brightness in the background regions, as shown in Fig. 3. In AECCA framework, it not only preserves the lighting information very well but also produces a natural scene effect. Also, the texture details are well preserved in the proposed image compared to all comparable methods without noise.

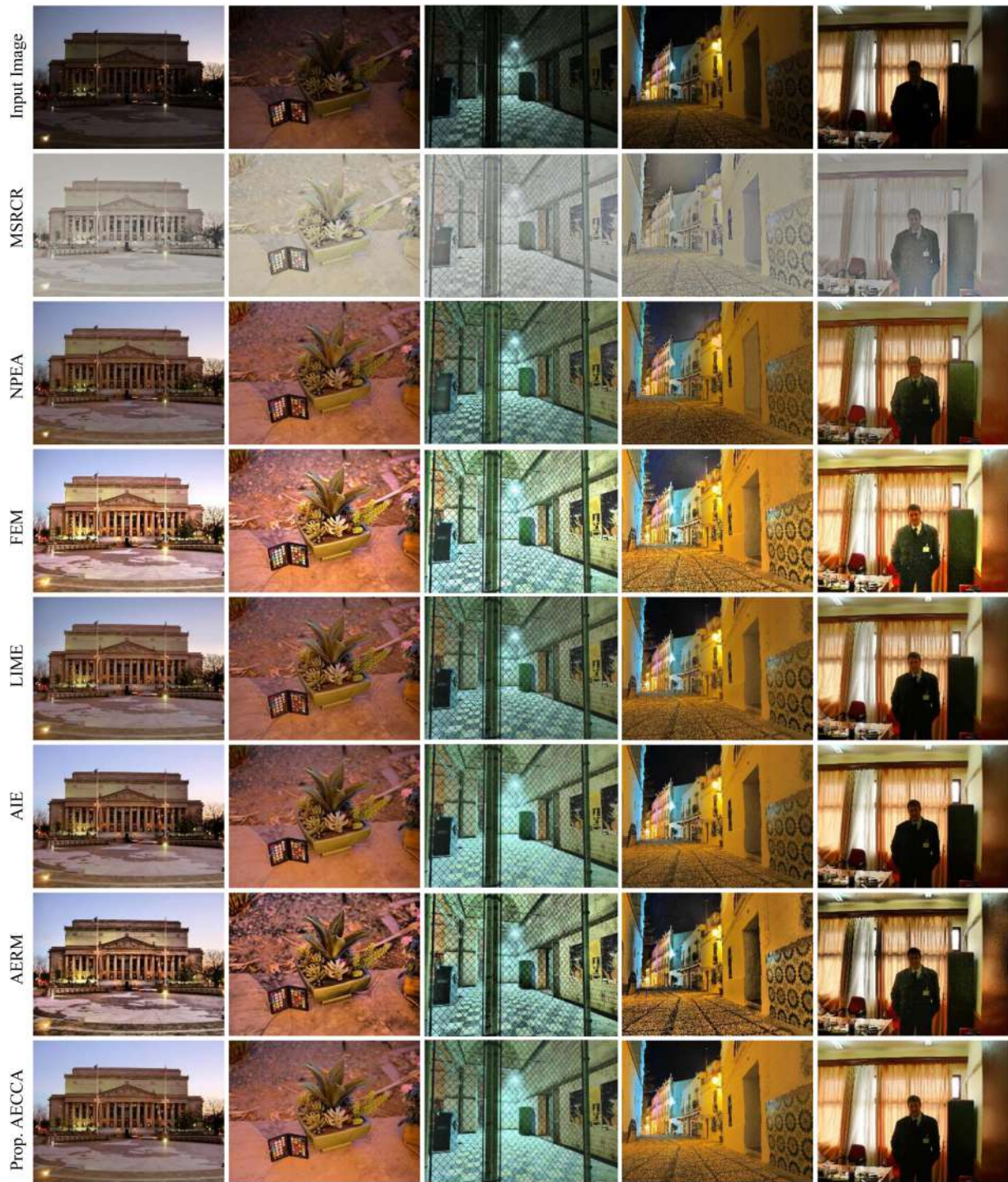


Fig. 6. Input low light image and enhanced images after applying low light image enhancement methods.

The proposed method highlights the hidden information with natural color and brightness. All comparative low light image enhancement approaches generate insufficient contrast, unnatural brightness, unpleasing artefacts, and color distortion and amplify the noise for the weakly illuminated portion, as shown in Figs. 3–6. Conversely, the proposed method produces a pleasing result because the designed AECCA method has focused on all essential characteristics like contrast, color, brightness, texture,

and structural details by several different proposed functions. The CCA makes a proper combination of two proposed images where the first Retinex-based image ideally deals with structural, textual, and color details.

In comparison, high contrast and natural brightness of dark and bright regions are perfectly dealt with by the second adaptive hyperbolic secant distribution-based image in which illuminance's feature is used. The proposed method can be used in

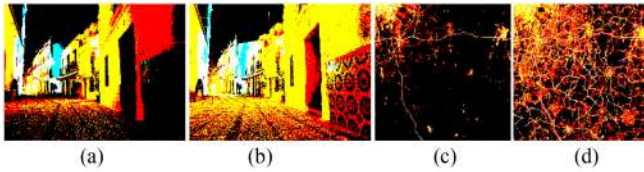


Fig. 7. Segmented images using Otsu method for low-light Input images and enhanced images.

TABLE I
AVERAGE VALUE OF QUANTITATIVE PARAMETERS OF LOW-LIGHT ENHANCED IMAGES FOR EACH METHOD

Methods	AME	SDME	NIQE	LOE	Time
MSRCR	18.190	30.412	3.181	888.049	1.397
NPEA	17.997	30.131	3.010	239.126	64.158
FEM	17.013	29.263	3.271	887.029	2.712
LIME	17.384	29.623	3.051	222.054	1.184
AIE	<u>18.693</u>	<u>30.423</u>	2.805	<u>80.268</u>	<u>1.551</u>
AERM	16.515	28.840	3.098	430.781	48.752
AECCA	18.999	30.821	<u>2.870</u>	48.0379	2.149

The best value is indicated in bold.

industrial applications because the segmented regions of the image are essential and utilized in intelligent driving, object detection, and many computer vision systems. The segmented images based on the Otsu method for low-light images and their AECCA-based enhanced images are shown in Fig. 7. It is clearly observed that the image's regions are finely segmented in enhanced based images, as shown in Fig. 7(b) and (d), compared to without enhanced image, as mentioned in Fig. 7(a) and (c).

B. Objective Evaluation

The average value of the AME, SDME, NIQE, LOE, and computational time (Time) comparable parameters for all 90 dark images are mentioned in Table I, and the best value is indicated as bold faces. The second-best value is denoted as underlined faces. The AME and SDME are the enhancement metrics, which measure the local contrast enhancement; the larger value of AME and SDME defines better enhancement performance. AME deals with the enhanced details using the relationship between the sum of the two luminance and the spread values observed in a window. SDME computes the change ratio of the variation speed of intensity values of pixels by the concept of the second derivative. The NIQE is highly associated with human perception, and it computes the distortion of the image without assuming any specific distortion.

The LOE uses enhanced and input images to obtain the lightness order error; generally, it refers to the lack of preservation of lightness order. So, the lower NIQE and LOE values prefer better enhancement performance for low-light images. The proposed AECCA framework shows the best average values for AME, SDME, and LOE, whereas it demonstrates the second-best value for NIQE. So, the presented method generates the best results regarding local contrast with hidden details and the ratio of

variation speed pixel values and lightness order. It also produces the second-best result for human perception function from a minimal margin.

The computation time (seconds) of the proposed method is slightly higher than the MSRCR method, so computational cost and NIQE may be considered as the limitations of AECCA.

IV. CONCLUSION

A new CCA-based adaptive image enhancement technique is presented here for low-light images in which two proposed enhanced images are combined using the coefficients canonical correlation technique. First adaptive enhancement is obtained by the Retinex model, where the weighted multiscale Retinex model and adaptive enhanced reflectance image are designed. At the same time, the second adaptive enhanced image is generated by the beta-hyperbolic secant distribution function; in this article, again modified the logarithmic image processing model with the illuminance's factor. The introduced framework uses CCA for image fusion purposes first time in this article. Eight samples of low light images and their enhanced results of MSRCR, NPEA, FEM, LIME, AIE, and proposed AECCA frameworks are examples for subjective analysis. The visual evaluation of all methods concludes that the proposed method improves the image's brightness, contrast, and color naturalness without creating any artefacts. Different metrics such as AME, SDME, NIQE, LOE, and computation time are considered for the quantitative evaluation of dark images from standard datasets. NIQE and computational cost may be the limitation of the article because AECCA has slightly higher values than optimal values. Overall, the presented AECCA framework generates the best-enhanced results for weakly illuminated images for both subjective and objective assessments. So, the proposed low light image enhancement technique can benefit several computer vision and industrial applications, such as object recognition and tracking in weakly illuminated environments.

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